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Patient Safety Protocol from Pulmonary Perspective

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“This is the era of Evidence-based Medicine and patient safety should be a priority in healthcare practice.”

Effective communication between patient and healthcare team, practising evidence-based medicine, being ethical, constant learning and educating, capacity to deal with adverse events with a problem-solving attitude, and creating safe working environment are the pillars to build the basic foundation of patient safety [1].

From Pulmonary perspective, patient safety measures have to be taken at different levels involving different stakeholders. Actions are needed from patients and their caretakers, treating healthcare team, and administrative authorities.

The respiratory patients can actually behave either as source of infections or vulnerable to the airborne transmissible or droplet infections. The lungs are the most vulnerable organs as they are in direct contact with the environment during the process of respiration. As a result, any deleterious respirable elements ($<5\mu\text{m}$) in the environment can get access into the lower airways and the lungs via the upper airways. The acutely infected respiratory patients can be infectious to others while talking, coughing, and sneezing. On the contrary, the chronic respiratory patients are prone to harbour lung opportunistic infections from the environment and other infected patients. Patients can follow some very simple day-to-day non-pharmacological practices that can enhance and improve their quality of life. During the recent SARS-CoV-2 pandemic, such non-pharmacological interventions saved many lives worldwide.

I. Actions to be taken by patients

(A). Safety protocol for infectious respiratory patients

These patients should be careful so as not to transmit their infection to other healthy people or chronically ill patients.

1. Patients suffering from seasonal flu or other bacterial infections, preferably should wear a properly fitting clean face mask covering the nose and mouth correctly so that they do not spread the infectious microorganisms (virus and bacteria) to others. 2 ply cloth mask with increased thread count are non-inferior to surgical or N95 masks [5,6].

2. Such patients should follow the basic talking, coughing, and sneezing etiquettes. They should not shout and talk loudly and cover the nose and mouth with a piece of cloth or their hands or elbows while coughing or sneezing.

3. Patients should also be careful in disposing their infected sputum in proper dedicated disposing area in public places. At home, they can pour boiling water in the sputum container, spit in a bactericidal container or spit on a piece of paper or cloth and burn it immediately.

4. Patients should maintain physical distance of atleast 2 metres ($\approx$ 6 feet), while they are sick and infectious, both at home and work place [7].

5. Patients should maintain proper hand hygiene and be careful enough not to touch or rub their eyes with infected hands. They should wash their hands with soap and water frequently before and after touching objects.

6. Such patients should not share their belongings with others while they are sick, especially food items, cosmetics and beauty accessories (in case of females), comb, pen, etc.

7. Patients should also maintain proper upper airway hygiene by cleaning their nasal cavities, oral cavity, and throat regularly with antiseptic gargles or lukewarm saline water, early morning and preferably after each meal.

8. Such patients may preferably restrict themselves to home till the incubation period of their infection is over.

(B). Safety protocol for chronic respiratory patients

The chronic respiratory patients often have structural lung abnormalities due to which they are prone to harbour opportunistic microorganisms or any other microorganisms very easily in the lungs that lead to periodic exacerbations of their disease. Therefore, such patients should be very particular in following the basic health protocols, especially infection prevention practices in order to improve their quality of life. Such patients should practise the following-

1. Regular use of properly fitting face mask covering nose and mouth correctly, especially while moving outside in public and crowded places and hospital areas.

2. Maintain proper physical distance of atleast 2 metres ($\approx$ 6 feet) from infected people.

3. Maintain proper upper airway hygiene as mentioned in earlier section.

4. Maintain proper hand hygiene by washing with soap and water after touching infected objects.

5. Patients should avoid exposure to triggering factors that may aggravate their adverse lung conditions like organic and inorganic dust, smoke, inhalation of molds and allergens, excessive cold weather, and damp areas.

6. Such patients should maintain a high protein diet to replenish the chronic protein loss through their excessive sputum production (provided liver and kidney functions are normal). They should avoid stale and cold food items.

(C). Precautions to be taken during sputum collection

The respiratory patients should maintain utmost care while coughing up of own sputum for collection to test or discard so as not to cause health hazards to others in the vicinity. During coughing sputum, patients should choose an isolated area, whether indoor or outdoor. If indoor, windows should be wide open to the outside atmosphere, provided there is no one in the vicinity.

II. Role of treating healthcare team

Patient education

The treating team has to be active in properly counselling and educating the patients and caretakers regarding safety measures and ways of dealing with their disease that should include the following-

1. Proper counselling and education of patients regarding airborne infections and necessary precautions to be taken.

2. Avoidance of triggering factors (allergens, irritants, smoke, damp areas, cold exposure, etc) in case of asthmatic and chronic respiratory patients.

3. Proper inhalation techniques and care of drug delivery devices like spacer, nebulizer, etc.

4. Importance of drug compliance or adherence and education on adverse effects of drugs.

5. High protein diet for replenishing the constant protein loss in case of chronic sputum production (provided hepatic and renal functions are normal).

6. Regular use of properly fitting face mask as mentioned in earlier sections.

7. Maintenance of proper upper airway hygiene.

8. Maintenance of a diary or note book for self-monitoring of symptoms and exacerbations.

9. Recognition of red flag signs for health care visit.

10. Teaching family members or caretakers the different airway clearance techniques

like postural drainage in case of chronic respiratory patients with structural lung abnormalities.

11. Motivational and psychosocial counselling including options for meditation and vocational training.

12. Emphasize on smoking and tobacco cessation.

III. Role of Administrative Authority

Building proper Health Infrastructure

Ideally there should be a separate department of Infectious Disease management that takes care of all infectious and communicable diseases under one roof. It becomes easier to provide the right infrastructure to handle such cases. Whether outpatient department (OPD) or inpatient department (IPD), infectious and communicable disease patients cannot be managed under centralised air-conditioned system. The following should be kept in mind during building up of infrastructure for such patients-

1. A separate independent hospital or building with proper cross ventilation system (6-12 air changes per hour) in OPD and IPD blocks [1].

2. Separate isolation wards with negative pressure ventilation to manage the infectious patients within incubation period.

3. Provision of intensive care facilities in the IPD block to manage the critically ill patients.

4. Provision of a pulmonary rehabilitation centre in the building to manage the chronic patients, facilitated with a multidisciplinary team comprising of treating physician, physiotherapist, psychiatrist, dietician, and nursing personnel.

5. Virus-laden exhaust air should be treated with HEPA/UV (High-Efficiency Particulate Air Filter/Ultraviolet) equipment, where HEPA filters can capture particles as small as 0.3 microns, while UV light and TEG (triethylene glycol) can kill viruses and bacteria [4].

Challenges

There are various challenges in meeting all necessary requirements in the creation of patient safety environment. These can exist both at community and institutional levels [3].

In the community, social behaviour like lack of talking, coughing, and sneezing etiquettes, indiscriminate spitting habits, addictions of toxic substances, lack of proper

health education, etc, special populations like migrants, illiterate tribal and non-tribal communities, people residing in remote underdeveloped areas, high risk centres like old age homes, orphanages, mental asylums, prisons, lack of social support, etc, environmental factors like indoor and outdoor air pollution, are important issues to be dealt with seriously. At institutional level, lack of dedicated infectious disease management hospitals or buildings with proper isolation wards with negative pressure ventilation and intensive care facilities, overcrowding of patients and attendants in hospitals with lack of restricted entries, lack of trained manpower to deal with patient population in a systematic manner, centralised air-conditioned systems in hospitals without facilities of proper cross ventilation, lack of a robust system to segregate infectious and non-infectious patients, etc are matters of serious concern.

Recommendations

1. Enactment of effective legislation

Certain Laws or Acts should be passed and made mandatory by the Central and State Governments like compulsory free elementary education for all citizens of the country, penalisation for non-compliance to safe health practices, enforcement of strict civic rules for good health practices, complete ban of smoking and tobacco in any form, strengthening the pollution control laws of the country, etc. Influenza and pneumococcal vaccinations can be made mandatory for high-risk patients.

2. Utilization of social media

Local languages should be used more while utilizing social media for creating health awareness amongst people residing in remote and distant areas. Patients have to be properly educated in their language on multidisciplinary safe health practices and emphasized on strict compliance to those practices for their health benefit.

3. Establishment of separate hospitals for infectious disease management

There should be separate branch for Infectious Disease management and Government should build separate well-equipped hospitals to handle such diseases. That way it is easier to manage and focus on research related to such diseases.

To conclude, the ancient renowned Greek philosopher and polymath Aristotle propagated the principle of “mens sana in corpore sano” which means “a sound mind in a sound body”. Without a healthy body, the mind cannot function properly and in the absence of both, one may lead a miserable and piteous life dependant on others, contributing to the economic burden of the country. Hence, health and thereby patient safety measures should always be a priority for every citizen of a Nation.

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